EMMA REY SÁNCHEZ

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SUMMARY

AI Engineer at Deloitte with a BSc in Mathematical Engineering with Artificial Intelligence, currently pursuing an MSc in Advanced Artificial Intelligence. I combine mathematical reasoning with software development to build AI applications, with experience spanning model implementation, evaluation and integration into working systems.

EXPERIENCE

Artificial Intelligence Engineer **Sep 2026 - Present**
Deloitte Barcelona, Spain

- Developed a Retrieval-Augmented Generation (RAG) system for context-informed LLM responses.
- Contributed to computer vision and AI agent projects for the pharmaceutical industry.

EDUCATION

MSc in Advanced Artificial Intelligence **Sep 2025 - Dec 2026**
Universidad Pontificia Comillas ICAI Madrid, Spain

- Relevant modules: Deep Generative Models, Information Geometry, Probabilistic AI, Deep Reinforcement Learning.

Academic Exchange **Jan 2025 - Jun 2025**
University of Miami Miami, USA

- Engineering and AI, with business coursework. Grade: 9.4/10.

BSc in Mathematical Engineering with Artificial Intelligence **Sep 2021 - Jun 2025**
Universidad Pontificia Comillas ICAI Madrid, Spain

- Relevant modules: NLP, Deep Learning, Machine Learning, Computer Vision, Robotics and Reinforcement Learning.

A-Levels and GCSEs **Sep 2016 - Jun 2021**
Kew House School London, UK

- A-Levels: A*, A*, A*, A*, A* (Maths, Further Maths, Physics, Chemistry, Spanish).

COMPLEMENTARY EDUCATION

Diplomas in Personal and Professional Skills, Communication Skills and Foreign Language Studies
Universidad Pontificia Comillas

Sailing and Windsurfing Instructor Course | Escuela de Vela Socaire | Jun 2021

KEY ACHIEVEMENTS

Bachelor's Thesis: Hierarchical Music Genre Classification
Improved deep learning accuracy on the FMA dataset using two-stage training with macro-genres and NLP-based music metadata enhancement.

GNN Rewiring Analysis: GitHub
Compared four graph-rewiring strategies across three datasets, analysing over-squashing and over-smoothing with GAT models and graph diagnostics.

Bayesian Sentiment Analysis: GitHub
Co-developed a sentiment classifier using Bayesian mixture-of-experts, topic modelling and uncertainty estimation; evaluated calibration and out-of-distribution detection.

Image Captioning Project: GitHub
Implemented an InceptionV3-LSTM model with greedy and beam-search decoding, evaluated with BLEU and CIDEr.

Second Place in Algebra, Mathematics Team Contest, Comillas ICAI

ADDITIONAL INFORMATION

- **Languages** | Spanish (Native) | English (Bilingual; five years in London) | French (B2).
- **Technical Skills** | Python, SQL, PyTorch, scikit-learn, Hugging Face, MATLAB, R, Docker, MLOps, CI/CD, FastAPI, React, TypeScript, MongoDB, LLMs, RAG and AI Agents.
- **Teaching** | Volunteer English teacher in Trujillo, Peru (Jun-Jul 2024); sailing instructor at Escuela de Vela Socaire, Murcia (five summer seasons).